Surface Chemistry

Adsorption, absorption, sorption

Adsorption: Adsorption is the adhesion of atoms, ions or molecules from a gas or liquid on a solid surface. It is a surface phenomena.

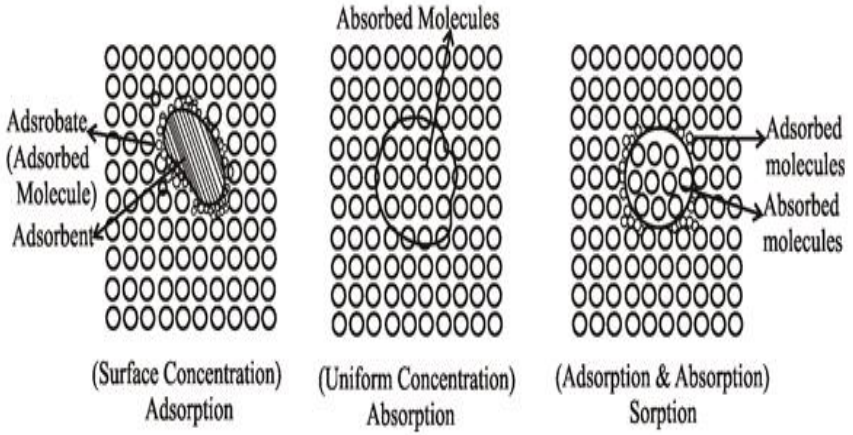
- The substance that deposits on the surface is called **adsorbate**
- and the solid on whose surface the deposition occurs is called the **adsorbent.**

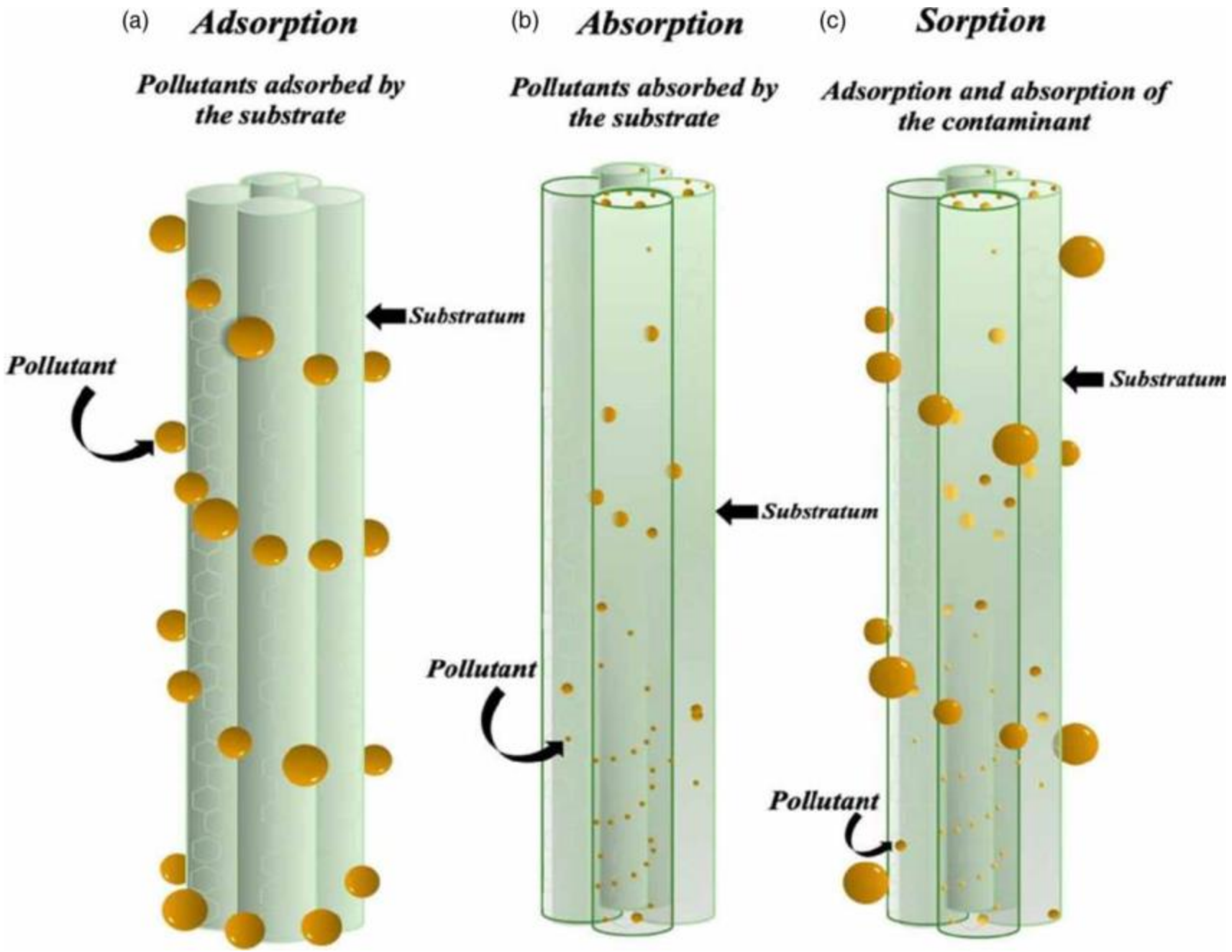
Absorption is a bulk phenomena in which the substances uniformly distributed throughout the body of solid or liquid to form a solution.

Sorption: The process in which both adsorption and absorption take place simultaneously is generally known as sorption.

Examples:

- i) The uptake of gases by zeolites.
- ii) The uptake of H_2 by charcoal.




(a) Adsorption
Pollutants adsorbed by the substrate
Substratum
Pollutant
(b) Absorption
Pollutants absorbed by the substrate
Substratum
Pollutant
(c) Sorption
Adsorption and absorption of the contaminant
Substratum
Pollutant

Difference between adsorption and absorption

Absorption	Adsorption
It is the process by which atoms, molecules or ions enters the bulk	The accumulation of the molecular species at surface not in the bulk
It is a bulk phenomenon	It is a surface phenome
It is an endothermic process	It is an exothermic process
Concentration is the same throughout the material	In this concentration at the surface is different than that of bulk of absorbent
It is not affected by temperature	It is temperature dependent phenomenon
Examples: i) water vapours absorbed by anhydrous $CaCl_2$. ii) NH_3 absorbed by water to form NH_4OH.	Examples: i) water vapours adsorbed by silica. ii) NH_3 adsorbed by charcoal.

Examples of adsorbents

- Silica gel, charcoal, metals (Ni, Cu, Ag, Pd, Pt etc.)

Examples of adsorbates

- There are various gases (H_2, N_2, O_2, Ne, SO_2, NH_3 etc.)

Factors affecting adsorption

- Adsorbent
  - A larger surface area of the adsorbent allows for more molecules to be adsorbed, leading to a greater adsorption capacity
- Adsorbate
  - Generally, substances with lower solubility in the surrounding medium are more readily adsorbed
  - The size and shape of the adsorbate molecules influence their ability to fit into the adsorbent's pores and interact with its surface
- pH
  - Affects surface charge of the adsorbent and the charge on the solute
  - Lowering pH increases adsorption for the organic materials
- Temperature
  - Exothermic process - hence increasing temperature decreases adsorption
- Pressure
  - Pressure leads to increase of adsorption and decrease of pressure causes desorption

Types of adsorption

There are two types of adsorption:

a) physical adsorption or physisorption and

b) chemical adsorption or chemisorption

a) **Physisorption**: This is due to the gas molecules being held to the solid surface by van der Waal's attractive forces. It is also referred to as van der Waal's Adsorption.

For example, adsorption of hydrogen or oxygen on charcoal is physisorption.

b) **Chemisorption**: In this kind of adsorption, the gas molecules or atoms are held to the solid surface by chemical bonds. These bonds may be covalent or ionic in nature.

For example, hydrogen is chemisorbed on nickel. Hydrogen molecules is first adsorbed by van der Waal's forces and then dissociates.

Difference between physisorption and chemisorption

Physisorption	Chemisorption
1. Low heat of adsorption usually in the range of 20-40 kJ mol^{-1}	High heat of adsorption in the range of 40-400 kJ mol^{-1}
2. Force of attraction are Van der Waal's forces	Forces of attraction are chemical bond forces
3. It usually takes place at low temperature and decreases with increasing temperature	It takes place at high temperature
4. It is reversible	It is irreversible
5. It is related to the ease of liquefaction of the gas	The extent of adsorption is generally not related to liquefaction of the gas
6. It is not very specific	It is highly specific
7. It forms multi-molecular layers	It forms monomolecular layers
8. It does not require any activation energy	It requires activation energy

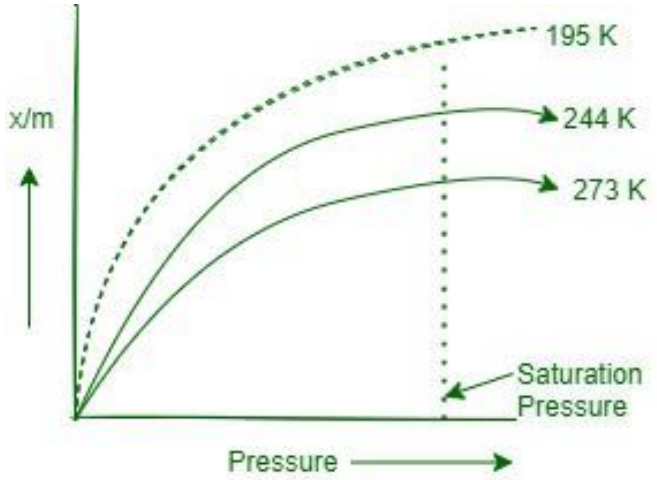
Adsorption isotherm

A graph between the amount of the gas adsorbed per gram of the adsorbent and the equilibrium pressure of the adsorbate at constant temperature is called the adsorption isotherm.

Adsorption Isotherm is a curve that expresses the variation in the amount of gas adsorbed by the adsorbent with pressure at constant temperature.

The plot between extent of adsorption (x/m) vs pressure at constant temperature.

Here, x = Mass of gas adsorbed
m = Mass of adsorbent



Types of physical adsorption isotherms

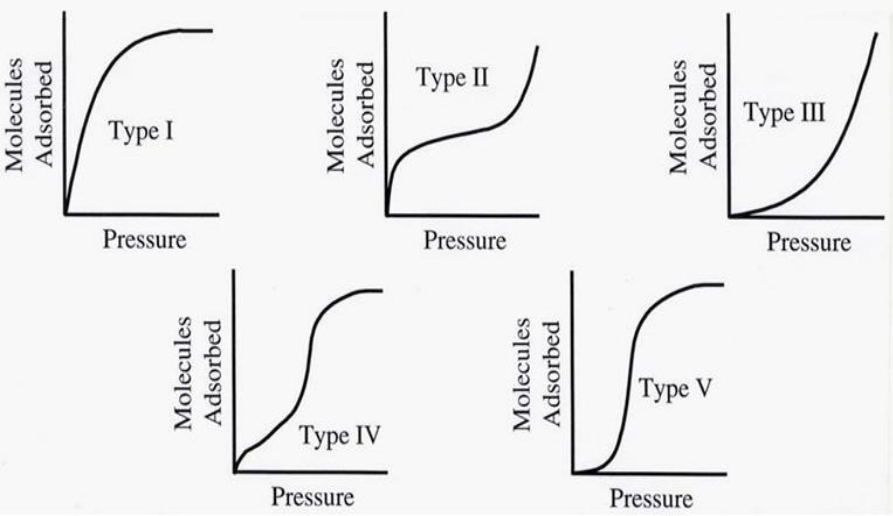
There are five general types of physical adsorption isotherms have been reported.

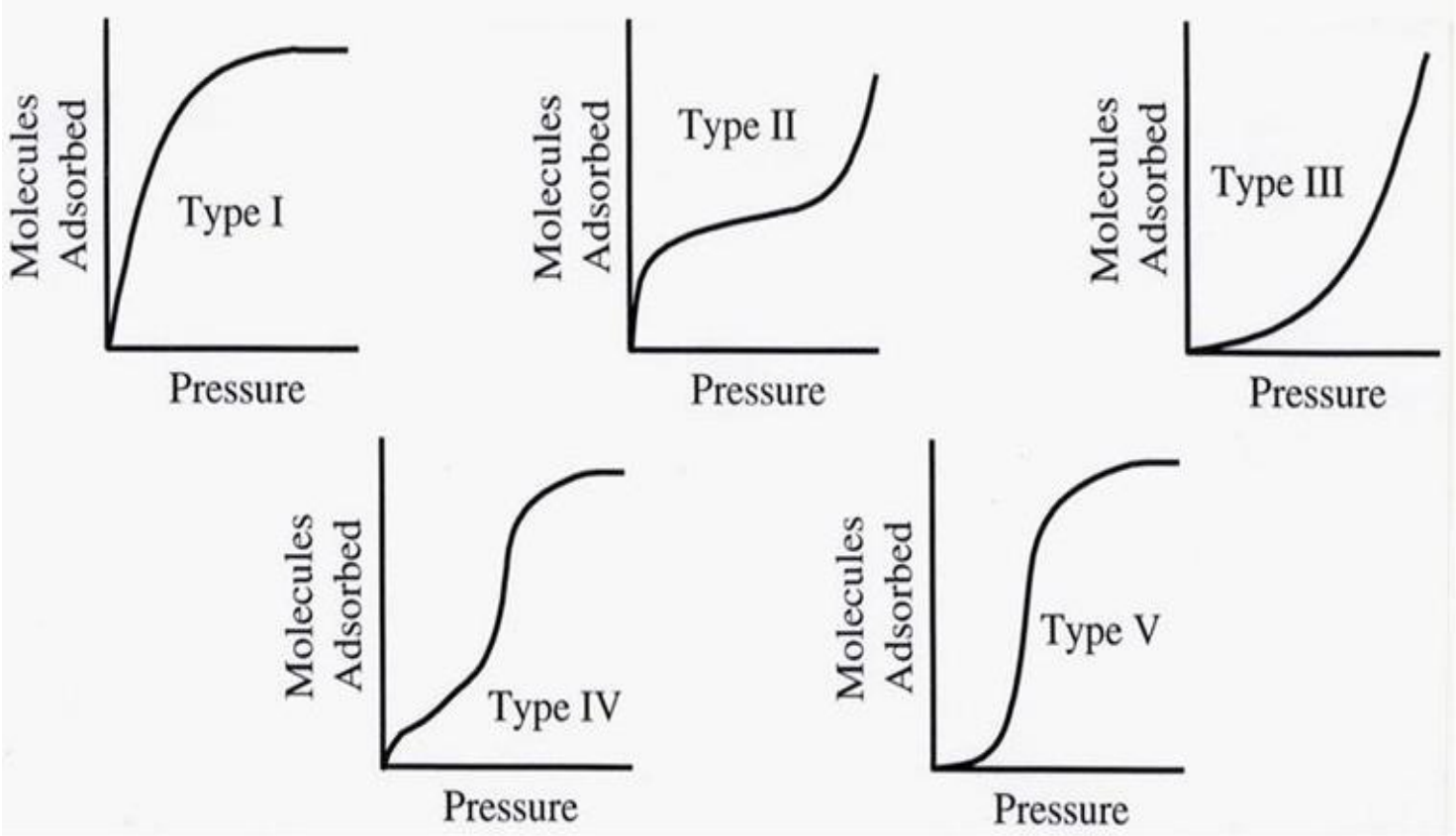
- **Type-I:** This type of curve is obtained in such cases where mono-molecular layer is formed on the surface of the adsorbent.

Examples: Adsorption of N_2 on charcoal at -195^0C.

- **Type-II:** This type of curve has a transition point which represents the pressure at which the formation of mono-molecular layer is complete and that of the multi-molecular layer is being started.

Examples: Adsorption of N_2 on silica gel at -195^0C.




Molecules Adsorbed
Type I
Pressure
Molecules Adsorbed
Type II
Pressure
Molecules Adsorbed
Type III
Pressure
Molecules Adsorbed
Type IV
Pressure
Molecules Adsorbed
Type V
Pressure

- **Type-III:** In this type of isotherm there is no transition point. In this type, multi-molecular layer formation starts even before the formation of mono-molecular layer is complete.

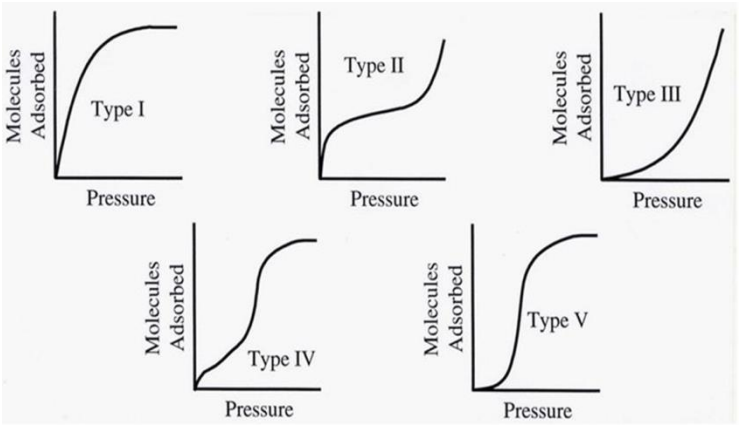
Examples: Adsorption of Br_2 or I_2 vapours on silica gel at 79^0C.

- **Type-IV:** In this case there is a tendency for a saturation state to be reached in the multi-molecular region as well. In fact this can be regarded as a duplication of the type-II.

Examples: Adsorption of benzene vapours on ferric oxide gel at 50^0C.

- **Type-V:** This isotherm indicates multi-molecular layer formation in the beginning. At higher pressure, there is a tendency for adsorbates (molecules adsorbed) to remain constant. It means that the saturation state has been reached.

Examples: Adsorption of water vapours on charcoal at 100^0C.



Adsorption isotherms

There are various types of adsorption isotherms

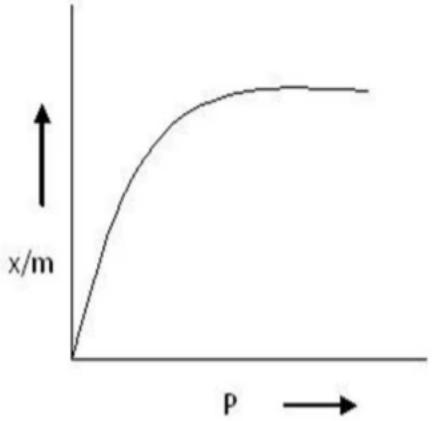
1) Freundlich adsorption isotherm
2) Langmuir adsorption isotherm
3) BET adsorption isotherm

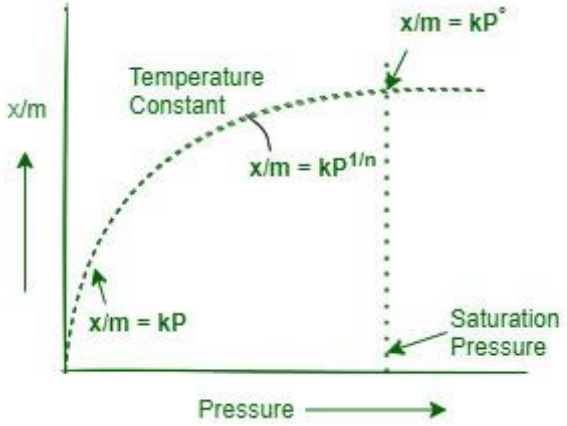
Freundlich isotherm

- Freundlich proposed an empirical relation in the form of a mathematical equation.

$$x/m = kP^{1/n}$$

- where x is the mass of the gas adsorbed on a mass m of adsorbent at a pressure P, k and n are constants depending on the nature of the gas and the adsorbent and on temperature.
- This relation is generally represented in the form of a curve obtained by plotting the mass of the gas adsorbed per unit mass of adsorbent (x/m) against equilibrium pressure.

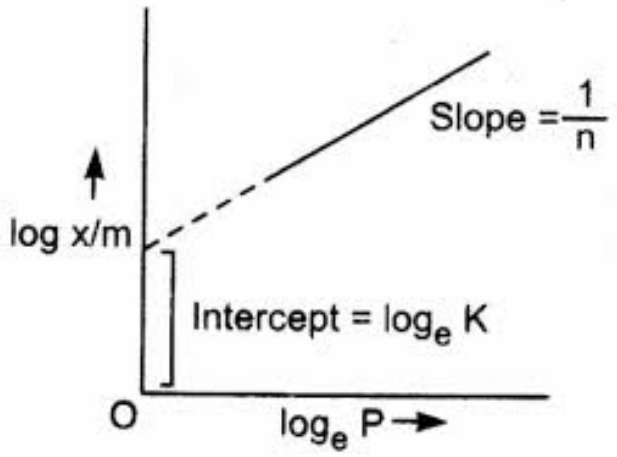




Freundlich isotherm is not applicable at high pressures. Taking logarithms on both sides of Freundlich equation, we have

$$\log(x/m) = (1/n)\log P + \log K$$

This equation is for a straight line.Thus a plot of log x/*m* against log *P* should be a straight line with slope 1/*n* and intercept log *k*.



However, it is actually found that the plots were straight lines at low pressures, while at higher pressure they showed a slight curvature, especially at low temperature.

This indicated that Freundlich equation is approximate and does not apply to adsorption of gases by solids at higher pressures.

Limitations:

1) It is a empirical formula without theoretical foundation.

2) It is valid over a certain range of pressure.

3) The constant k and n vary with temperature.

Langmuir isotherm's

Langmuir derived a simple adsorption isotherm based on theoretical considerations. It was named after him.

Assumptions

1) The layer of the gas adsorbed on the solid adsorbent is one-molecule thick.
2) The adsorbed layer is uniform all over the adsorbent.
3) The adsorption is taking place on the fixed sites and there is no interaction between the adjacent adsorbed molecules.
4) The process of adsorption is a dynamic process which consists of two opposing processes:

a) condensation and b) evaporation process.

Condensation process :It involves the condensation of gas molecules on the surface of the adsorbent.

Evaporation process :It involves the evaporation of gas molecules from the surface of the adsorbent.

When the equilibrium is reached, the rate of condensation becomes equal to the rate of evaporation.

Derivation of Langmuir Isotherm

Langmuir considered that the gas molecules strike a solid surface and are thus adsorbed. Some 'desorbed' fairly rapidly.

A dynamic equilibrium is eventually established between the two opposing processes, adsorption and desorption.

If θ is the fraction of the total surface covered by the adsorbed molecules, the fraction of the naked area is

$$(1 - \theta)$$

The rate of desorption (R_2) is proportional to the covered surface θ. Therefore,

$$R_2 = k_2\,\theta \qquad (1)$$

where k_2 is the rate constant for the desorption process.

The rate of adsorption (R_1) is proportional to the available naked surface $(1 - \theta)$ and the pressure (P) of the gas.

$$R_1 = k_1(1 - \theta)P \qquad (2)$$

where k_1 is rate constant for the adsorption process.

At equilibrium the rate of desorption is equal to the rate of adsorption. That is,

$$K_1\,(1-\theta)P = K_2\,\theta$$

$$K_1P - K_1\theta P = K_2\,\theta$$

$$K_1P = \theta\,(K_2 + K_1P) \qquad (3)$$

$$\theta = \frac{K_1 P}{K_2 + K_1 P}$$

$$\theta = \frac{(K_1 P)/K_2}{(K_2 + K_1 P)/K_2}$$

$$\theta = \frac{bp}{1+bp}, \text{ where } b = \frac{K_1}{K_2} \qquad \text{-----4}$$

The amount of the gas adsorbed on the surface of adsorbent is proportional to θ.

$$\frac{x}{m} \propto \theta$$

$$\frac{x}{m} = K_3\theta \ldots\ldots\ldots\ldots\ldots\ldots 5$$

Substituting the value of equation 4 , we get

$$\frac{x}{m} = K_3 \frac{bp}{1+bp}$$

$$\frac{x}{m} = \frac{ap}{1+bp} \ldots\ldots\ldots\ldots 6$$, where $a = K_3 b$ is constant.

Thus, Equation 6 is the required Langmuir equation, and a & b are Langmuir constant

On rearranging the above Langmuir equation, a straight line equation should be obtained.

$$\frac{x}{m} = \frac{aP}{1+bP}$$

$$\frac{x/m}{P} = \frac{a}{1+bP}$$

$$\frac{P}{x/m} = \frac{1}{a} + \frac{b}{a}\,\mathrm{P}$$

On plotting P/(x/m) against P, we should get a straight line with a slope equal to b/a and the intercept 1/a.

Why in certain cases, the experimental curves of Langmuir isotherm are not straight?

This may be due to the following reasons

1) non-uniformity of surfaces

2) formation of multi-layers in some cases

3) partial adsorption

4) some chemical reaction of the adsorbate with the adsorbent.

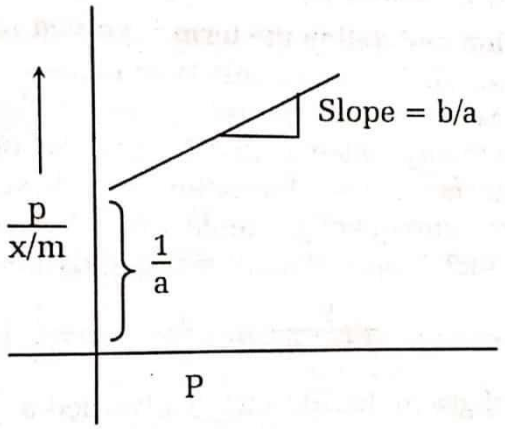


Fig. p/x/m Vs P plot

Limitations:

1) Langmuir postulated that a saturated value of adsorption is independent on the temperature. But experiment shows that it is actually falling off with rising temperature.

2) Langmuir assumed that an adsorption film on a plane surface will never be over a molecule thick. In actual practice, much thicker films have been reported.

3) This theory cannot explain the five types of adsorption isotherms.

BET isotherm

- BET theory aims to explain the physical adsorption of gas molecules on a solid surface and serves as the basis for an important analysis techniques for the measurement of the specific surface area of a material.
- In 1938, Stephen Brunauer, Paul Hugh Emmett and Edward Teller published an article on physical multi-layer adsorption theory in a journal for the first time.
- BET consists of the first initials of their family names.
- The concept of the theory is an extension of the Langmuir theory.
- The resulting BET equation is express by

$$\frac{1}{\left[V_a\left(\frac{P_0}{P}-1\right)\right]}=\frac{C-1}{V_m C}\times\frac{P}{P_0}+\frac{1}{V_m C}$$

$$\frac{1}{\left[V_a\left(\frac{P_o}{P}-1\right)\right]}=\frac{C-1}{V_mC}\times\frac{P}{P_o}+\frac{1}{V_mC}$$

P= partial vapor pressure of adsorbate gas in equilibrium with the surface at 77.4 K (b.p. of liquid nitrogen), in Pa,

P_o= saturated pressure of adsorbate gas, in Pa,

V_a= volume of gas adsorbed at standard temperature and pressure (STP) [273.15 K and atmospheric pressure (1.013 × 105 Pa)], in mL,

V_m= volume of gas adsorbed at STP to produce an apparent monolayer on the sample surface, in mL,

C= dimensionless constant that is related to the enthalpy of adsorption of the adsorbate gas on the powder sample.

- In the BET equation, V is the volume of gas adsorbed at pressure P and temperature T, V_m is the volume of gas corresponding to monolayer coverage, P^0 is the saturated vapour pressure of the adsorbate at temperature T and C is a BET constant and expressed by

$$C = exp\left(\frac{E_1 - E_L}{RT}\right)$$

- where E_1 is the heat of adsorption of the first layer and E_L is the heat of liquefication of the gas.
- The proposed four types (II - V) of isotherms for multi-layer can be explained as –

The isotherm of type II follow when $E_1 > E_L$ and for type III $E_L > E_1$.

The isotherm of type IV and V explain not only multi-layer adsorption but also condensation of gas in the pores and capillaries of the adsorbent.

- The types arise from the relative magnitude of E_1 and E_L .

The isotherm of type IV follow when $E_1 > E_L$ ..
Whereas it follow $E_L > E_1$ for type V.

$$\frac{1}{\left[V_a\left(\frac{P_o}{P}-1\right)\right]}=\frac{C-1}{V_mC}\times\frac{P}{P_o}+\frac{1}{V_mC}$$

- If we plot $P/V_a(P_0-P)$ vs. P/P_0, it showed be a straight line with slope $(C-1)/(V_mC)$ and intercept $1/V_mC$.
- From these slope and intercept V_m and C can be found.

Although this theory have proved fairly successful in explaining the more complex types of isotherms, they are still insufficient to account quantatively for all the phenomenon observed.

BET isotherm is not accurate at all pressure, but it widely used in industries to determine the surface areas of solids.

Heat of adsorption

- The heat of adsorption, which is the negative enthalpy of adsorption, can be obtained through simultaneous measurement of the adsorbed amount and the evolved heat.

- Calorimetric measurements of the evolved heat are mostly measured during vapor-phase adsorption.

- The main advantage of the calorimetric measurement is that it provides a direct and absolute measure of the ‘strength’ of interaction and a quantitative analysis of the site distribution.

- The obtained enthalpies can be compared to the results of calculations or other parameters.

Gibbs adsorption equation

- Gibbs adsorption equation is a thermodynamic expression which relates the surface concentration (or surface excess) of a species to both surface tension (γ) and the bulk activity or fugacity of the adsorbate.
- For liquid-liquid and liquid-vapour system the Gibbs adsorption equation may be used to determine the surface concentration and γ.
- For solid-gas system the Gibbs adsorption equation may be used to determine the surface concentration but surface tension cannot.

A general change in Gibbs energy (G) is brought about by changes in T, p and n_J

$$dG = -SdT + Vdp + \gamma d\sigma + \sum \mu_J dn_J \qquad (1)$$

where, σ is surface area, μ is chemical potential, J is number of components and n_J is overall amounts.

- When this relation is applied at G, G(α), G(β), that is the surface σ, bulk solution phase α, and the vapor phase β, we find

$$dG(\sigma) = -S(\sigma)dT + \gamma d\sigma + \sum \mu_J dn_J(\sigma) \qquad (2)$$

- Because at equilibrium the chemical potential of each component is the same in every phase,

$$\mu_J(\alpha) = \mu_J(\beta) = \mu_J(\sigma)$$

- Therefore the integration form of the equation (2) is given as

$$G(\sigma) = \gamma\sigma + \sum \mu_J n_J(\sigma) \qquad (3)$$

- At constant temperature equation (2) can be written as

$$dG(\sigma) = \gamma d\sigma + \sum \mu_J dn_J(\sigma) \qquad (4)$$

- Differentiation form of equation (3)

$$dG(\sigma) = \gamma d\sigma + \sigma d\gamma + \sum \mu_J dn_J(\sigma) + \sum n_J(\sigma) d\mu_J \qquad (5)$$

- From equation (4) and (5) we get

$$\sigma d\gamma + \sum n_J(\sigma) d\mu_J = 0$$

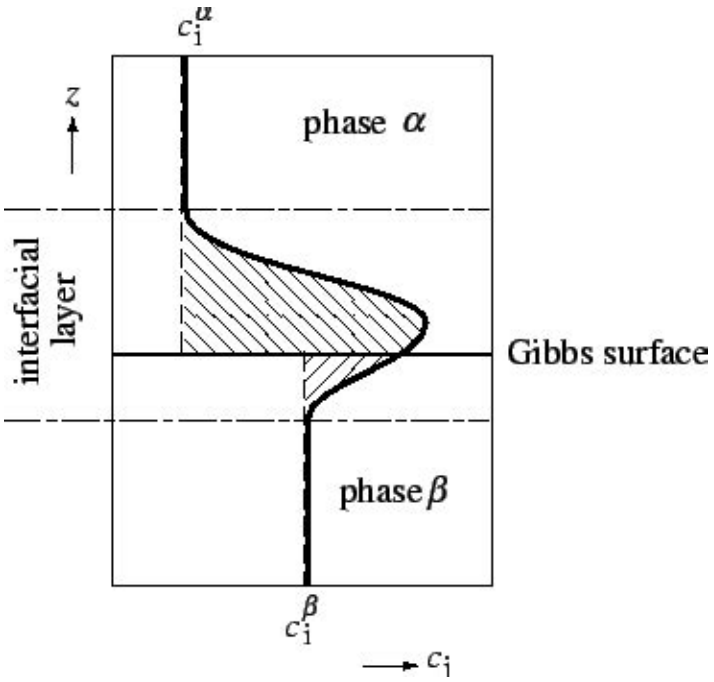
$$d\gamma + \sum n_J/\sigma(\sigma) d\mu_J = 0$$

$$- d\gamma = \sum \Gamma_J d\mu_J \qquad (6)$$

- where $n_J/\sigma(\sigma) = \Gamma_J$ and it is either the surface concentration or surface excess of component J.
- Equation (6) is a general form of Gibbs adsorption equation.

Surface excess

- It is a measure of the excess amount of a substance present at the surface of a system compared to the amount that would be present if the composition of the surface were the same as that of the bulk phases.



- We consider two phases α and β in contact and suppose that the system consists of several components J, each one present in an overall amount n_J.

- If the components were distributed uniformly through the two phases right up to the interface, which is taken to be a plane of surface σ, then the total Gibbs energy would be the sum of the Gibbs energies of both phases:

$$G = G(\alpha) + G(\beta)$$

- However, the components are not uniformly distributed because one may accumulate at the interface.

- As a result, the sum of the two Gibbs energies differ from G by an amount called the surface Gibbs energy G(σ): i.e.

$$G = G(\alpha) + G(\beta) + G(\sigma)$$

or,
$$G(\sigma) = G - \{G(\alpha) + G(\beta)\}$$

➢Similarly, if it is supposed that the concentration of a species J is uniform right up to the interface , then from its volume we would conclude that it contains an amount $n_J(\alpha)$ of J in phase α and an amount $n_J(\beta)$ in phase β.

➢Therefore, $n_J = n_J(\alpha) + n_J(\beta)$

- However, because a species may accumulate at the interface, then the total amount of J differs these two amounts by

$$n_J = n_J(\alpha) + n_J(\beta) + n_J(\sigma)$$

or, $$n_J(\sigma) = n_J - \{n_J(\alpha) + n_J(\beta)\}$$

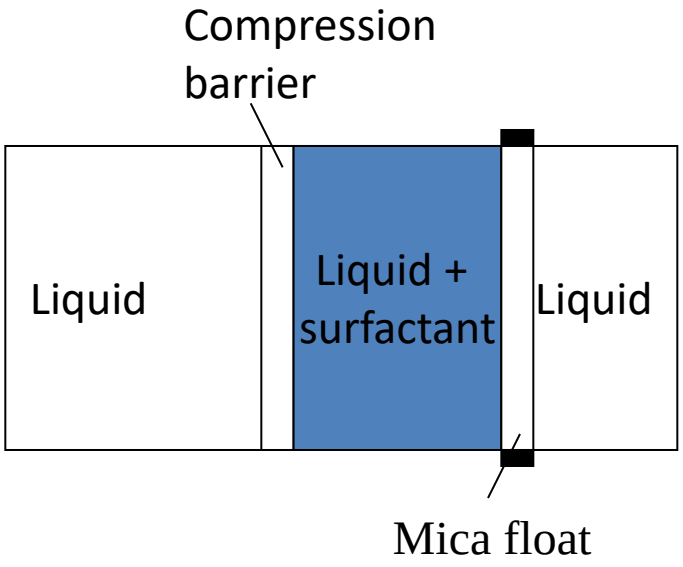
- This excess can be expressed as an amount per unit area of the surface by introducing the surface excess Γ_J;

$$\Gamma_J = n_J(\sigma)/\sigma$$

- Γ_J and $n_J(\sigma)$ may be either positive (an accumulation of J at the interface) or negative (a deficiency there).

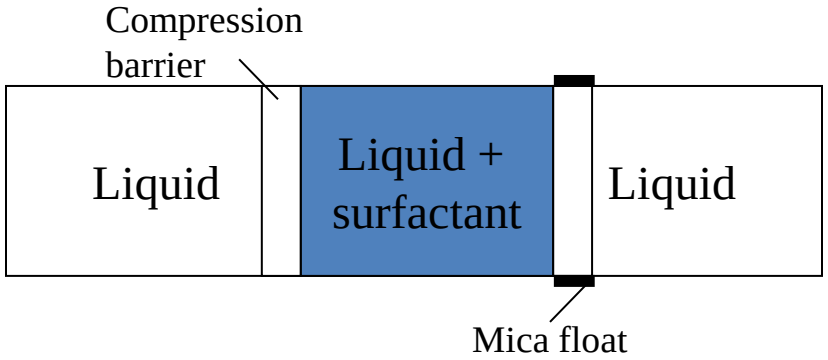
Surface films

- Surface film is the thin film of liquid near its surface and having the thickness equal to the molecular range of attraction for that liquid. The phenomenon of surface tension is associated intimately with this film.



Surface film balance

- The goal of surface film investigation is to establish the relation between the surface tension and the surface concentration of a monolayer of surfactant molecules, and hence the closeness of the surfactant molecules to one another on the surface.
- The device consists of a thin channel and a barrier that can be moved along the surface of the liquid in the channel, and hence compress any monolayer on the surface.
- The surface pressure (π) is the difference between the surface tension of the pure solvent and the solution ($\pi = \gamma^0 - \gamma$), it is measured by using a torsion wire attached to a strip of mica relax on the surface and against which one edge of the monolayer is pressed.
- In an actual experiment, a small amount (about 0.01 mg) of the surfactant is dissolved in a volatile solvent and then poured on to the surface of water;
- the compression barrier is then moved across the surface and the surface pressure exerted on the mica bar is monitored.



Applications of adsorption

1) Air pollution masks

- These consists of silica gel or activated charcoal powder, when dust or smoke are passed through them, particles get adsorbed on the surface of these materials.

2) Separation of noble gases by Dewar's flask process

- A mixture of noble gases of Ne, Ar, Kr is passed through Dewar's flask in presence of heated coconut charcoal. Ar and Kr gels are adsorbed leaving Ne.

3) Purification of water

- By the addition of alum stone to the water, impurities get adsorbed on the alum and water gets purified.

4) Removal of moisture

- Moisture in the air is removed by placing silica gel on which water molecule gets adsorbed.

5) Adsorption chromatography

- It is used to separate pigments and hormone.

6) Ion exchange method

- This method is used to remove hardness of water, calcium and magnesium ions get adsorbed on the surface of ion exchange resin

7) In metallurgy

- Froth floatation process is used for the concentration of metal ore, the particle gets adsorbed on the froth.